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Systematic review

Acupuncture for Parkinson's disease-related constipation: A systematic review and meta-analysis

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ABSTRACT

Introduction: The hazard of Parkinson's disease-related constipation (PDC) cannot be ignored. Acupuncture has the potential to become a mainstream treatment option for PDC. This review aimed to investigate the effect of acupuncture on PDC, which might provide evidence-based recommendations for clinical application.**Methods:** Six databases were searched for randomized controlled trials (RCTs) about acupuncture for PDC. The primary outcome was response rate. The secondary outcomes were rectal resting pressure, anal resting pressure, anal maximum contraction pressure, rectal sensation threshold, rectal maximum tolerable volume, the Constipation Scoring System (CSS) score, the Patient Assessment of Constipation Quality of Life (PAC-QOL) score, and adverse events (AEs).**Results:** Eight trials involving 591 participants were included. The methodological quality was generally low. A significant difference was found in response rate when comparing acupuncture with medicine (95 %CI= 1.09 to 1.54, $P = 0.003$) or blank control (95 %CI= 1.04 to 1.74, $P = 0.02$), while no significant difference was observed when comparing acupuncture + medicine with medicine. Acupuncture + medicine did have a significant effect on rectal resting pressure, anal resting pressure, anal maximum contraction pressure, rectal sensation threshold and rectal maximum tolerable volume when compared with medicine, while no significant effect was found when acupuncture was compared with medicine. A significant difference was found in the CSS score when acupuncture was compared with blank and in the PAC-QOL score when acupuncture was compared with sham or blank, while no significant difference was found in the PAC-QOL score when acupuncture was compared with medicine. One study reported mild AEs.**Conclusions:** Acupuncture might be an effective treatment for PDC. Due to the limited quality of evidence, high-caliber and multi-center RCTs with large sample sizes should be carried out. (PROSPERO ID is CRD42023422102)

1. Introduction

As one of the most common non-motor symptoms, constipation affects about 52 % of Parkinson's disease (PD) patients [1]. Parkinson's disease-related constipation (PDC) can occur in all stages of PD, and is correlated with the duration and severity of PD. It usually appears earlier than typical motor symptoms and worsens as PD progresses [2]. In recent years, constipation has been recognized as a risk factor for PD, which can increase the risk of developing PD [3]. PDC is also associated

with cognitive decline and may serve as a clinical biomarker for cognitive impairment [4]. Besides, PDC can reduce patients' quality of life, affect the absorption of dopaminergic drugs, and lead to false intestinal obstruction. Thus, the hazard of PDC cannot be ignored.

The pathophysiology of PDC is complex. A typical pathological characteristic of PD is the aberrant aggregation of α -synuclein within the central nervous system. α -Synuclein also begins in the submucosa from the enteric nervous system, and then travels retrogradely through the gut-brain axis to the central nervous system, thus causing motor

Abbreviations: PD, Parkinson's disease; PDC, Parkinson's disease-related constipation; RCTs, randomized controlled trials; PRISMA, the preferred reporting items for systematic reviews and meta-analyses; CNKI, the China national knowledge infrastructure; CSS, the constipation scoring system; PAC-QOL, the patient assessment of constipation quality of life; AEs, adverse events; RR, risk ratio; CI, confidence interval; MD, mean difference.

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symptoms of PD. Regarding the special mechanism of constipation in PD, degeneration of the vagus nerve and enteric nervous system occurs as a result of the accumulation of α -synuclein, which may lead to the dysregulation of intestinal fluid secretion and neurotransmitter function, thus further impeding intestinal motility and promoting PDC [5]. PD is also associated with the over-expression of pro-inflammatory cytokines in the gastrointestinal tract, which may lead to a weakened contraction of intestinal smooth muscles and promote the development of constipation [4]. Gut microbiota imbalance is often seen in PD patients, which affects the energy supply of colonocytes, the intestinal barrier and the intestinal inflammatory response [3]. Besides, anti-PD drugs, such as levodopa, ropinirole and piribedil, can promote an increased incidence of constipation [6]. Conventional treatments for PDC mainly include osmotic laxatives, stimulant laxatives, chloride channel activators, prucalopride, mosapride, lubiprostone, linaclotide, polyethylene glycol, lactulose, 5-hydroxytryptamine 4 receptor agonists, and so on [6]. While these treatments are often ineffective, because the pathophysiology of constipation is intermingled with PD and some drugs for PD can even induce and aggravate constipation [7].

Acupuncture, as an essential part of alternative medicine, has been widely used for PD and chronic constipation [8,9]. A number of articles published in recent years have suggested that acupuncture has the potential to become a mainstream treatment option for PDC in terms of increasing spontaneous bowel movements, enhancing motor function, and improving quality of life in patients with PD [10,11].

Until now, one systematic review investigating the effect of acupuncture for non-motor symptoms of PD, such as insomnia, depression, cognition, constipation and fatigue, has been published [12]. It only included four randomized controlled trials (RCTs) about the effect of acupuncture for PDC. Of the four included RCTs, two adopted Chinese patent medicine as the control method. These limitations rendered the evidence of acupuncture for PDC inconclusive. Since that time, some new RCTs of acupuncture for PDC have been published. In view of this, we conducted a systematic review and meta-analysis to investigate the effect of acupuncture on PDC, which might provide evidence-based recommendations for clinical application.

2. Methods

2.1. Protocol and registration

This review was registered with PROSPERO (CRD42023422102) and conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement checklist [13].

2.2. Search strategy

PubMed, Embase, CENTRAL, the China National Knowledge Infrastructure (CNKI), WanFang and VIP databases were searched for RCTs about acupuncture for PDC. The range of publication time was from the inception of each database to 12 April 2023. The following terms were used to search: ("acupuncture" OR "acupuncture therapy" OR "acupuncture, ear" OR "acupuncture points" OR "acupuncture analgesia" OR "acup*" OR "needl*" OR "point*") AND ("Parkinson Disease" OR "Parkinson" OR "parkinson*" OR "paralysis agitans") AND ("constipation" OR "dyschezia" OR "colonic inertia"). The detailed search strategies are shown in **Appendix A**.

2.3. Eligibility criteria

Inclusion criteria were listed as follows. (1) Study design: published RCTs. (2) Participants: patients diagnosed with PD and constipation. There were no restrictions on the age, gender or race. (3) Intervention: acupuncture alone, or acupuncture plus conventional treatment which was also applied in the control group. Acupuncture therapies using needles to insert into specific acupoints were included, such as manual

acupuncture, electroacupuncture, warm acupuncture, press-needle and so on. (4) Control: conventional treatment, sham acupuncture, or blank control. (5) Outcomes: the primary outcome included response rate, a recognized indicator for evaluating the efficacy of constipation. Response rate was counted by this formula: (number of cured patients + number of improved patients)/ total number of patients $\times 100\%$. The secondary outcomes included anal rectum dynamics (rectal resting pressure, anal resting pressure, anal maximum contraction pressure, rectal sensation threshold, and rectal maximum tolerable volume), the Constipation Scoring System (CSS) score, the Patient Assessment of Constipation Quality of Life (PAC-QOL) score, and adverse events (AEs).

Studies were excluded if they met the following criteria. (1) Studies including laser acupuncture, ear acupressure, transcutaneous electrical acupoint stimulation or moxibustion. (2) Studies with incomplete or inaccessible data. (3) Conference articles, theses or dissertations. For duplicate studies, only the most recent published was included.

2.4. Study selection and data extraction

Two reviewers (XXF and LC) independently evaluated eligible articles. After removing duplicates, these two reviewers screened the records by titles and abstracts, and then checked the full-text articles. The third reviewer (YYX) determined any divergence in the study selection process and ultimately confirmed the study selection result.

Two reviewers (XXF and LC) independently extracted the following information: first author, publication year, sample sizes, age of participants, intervention characteristics, and outcome measures. The third reviewer (YYX) determined any divergence in the data extraction process and ultimately confirmed the data extraction result.

2.5. Assessment of risk of bias

Two reviewers (XXF and LC) independently assessed the risk of bias by using the Cochrane Collaboration's tool [14]. Seven items (random sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data, selective reporting, and other bias) were judged as "low risk", "unclear risk", or "high risk". The third reviewer (YYX) determined any disagreements in the risk of bias assessment process and ultimately confirmed the assessment result.

2.6. Data analysis

RevMan 5.3 was utilized for statistical analysis. Risk ratio (RR) with 95 % confidence interval (CI) was used for dichotomous outcomes, and mean difference (MD) with 95 % CI was used for continuous outcomes. Heterogeneity was assessed using the I^2 test. Regardless of heterogeneity, a random-effects model was employed for meta-analysis. If sufficient trials (≥ 10 trials) were included, the funnel plot was used to assess publication bias. For studies with outcomes that could not be combined, we provided a description of each individual study. If the data extraction was insufficient for a meta-analysis, we conducted a qualitative analysis instead.

3. Results

3.1. Study selection

A total of 599 records were identified from PubMed ($n = 55$), Embase ($n = 49$), CENTRAL ($n = 69$), CNKI ($n = 259$), WanFang ($n = 110$) and VIP ($n = 57$). After removing duplicate records, the number decreased to 452. Then 437 records were excluded after screening titles and abstracts. The remaining 15 reports were assessed by checking the full text. Seven of them were excluded due to not RCTs ($n = 1$), ineligible population ($n = 5$), and no available data ($n = 1$). These seven reports excluded by checking the full text are listed in **Appendix B**. Finally, eight studies

were included [15–22]. A flow diagram of study selection is shown in Fig. 1.

3.2. Study characteristics

Table 1 lists the characteristics of the included studies. Eight studies with 591 participants were included, with 296 in the treatment group and 295 in the control group. 78 participants received acupuncture + medicine, 218 participants received acupuncture alone, 154 participants received medicine treatment alone, 39 participants received sham acupuncture, and 102 participants received a blank. Five studies adopted Chinese diagnostic criteria for PD in 2016 [15,16,18,19,22], two used UK Brain Bank criteria for PD [17,20], and one used the Movement Disorder Society's revised clinical diagnostic criteria for PD [21]. Seven studies adopted Rome III diagnostic criteria for constipation [15–20,22], and one study used Rome IV diagnostic criteria for constipation [21]. Overall, the average age of included participants was over 60. Five studies reported response rate [15,17,18,20,22]. Four studies reported anal rectum dynamics [15,17,18,20]. Of these four studies, three measured rectal resting pressure, anal resting pressure and anal maximum contraction pressure [17,18,20], all measured rectal sensation threshold [15,17,18,20], and three measured rectal maximum tolerable volume [15,17,20]. Two studies reported the CSS score [16,19], four studies reported the PAC-QOL score [16,19–21], and only one

described AEs [21].

3.3. Acupuncture regimen and control interventions

Three studies adopted electroacupuncture [15,17,18], two adopted warm acupuncture [16,19], two adopted press-needle [20,22], and the remaining one used manual acupuncture [21]. ST25 (*tianshu*), ST36 (*zusanli*) and ST37 (*shangjuxu*) were the most commonly used acupoints. The frequency of acupuncture ranged from twice a week to once a day. The total sessions ranged from 4 to 28. The course ranged from 7 to 30 days.

Four studies used conventional medicine treatments [15,17,18,20], such as phenolphthalein tablet [15,18], polyethylene glycol [17], and mosapride citrate tablet [20]. Three used blank control [16,19,22]. The remaining one adopted sham acupuncture as a control [21]. In this study, the sham acupuncture method was non-penetration at the acupoints [21].

3.4. Risk of bias

The risk of bias assessment is shown in Fig. 2 and Appendix C. Seven studies described sufficient details of randomization [15,17–22], and the remaining one only mentioned randomization [16]. Only two studies emphasized allocation concealment [20,21]. One study

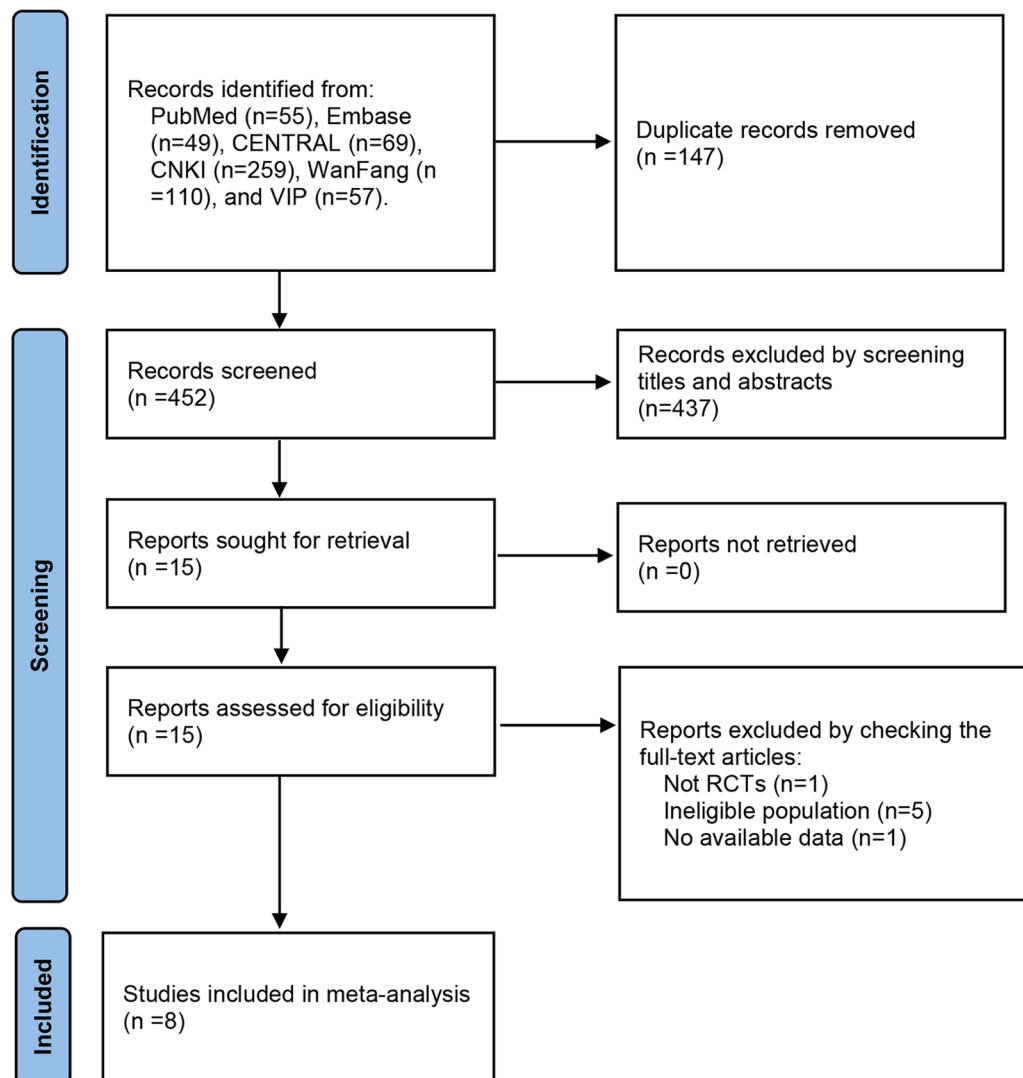


Fig. 1. Flow diagram of the study selection process.

Table 1

Characteristics of the included studies.

Study	Sample size (T/C)	Age (years)	Treatment group	Details of acupuncture	Control group	Outcomes
Luo et al. [15]	60 (30/30)	T: 66.31 ±5.57 C: 66.07 ±5.48	Electroacupuncture + Phenolphthalein tablet	ST36, ST25; continuous wave, 1.5 Hz; 20 min each time, once a day, total 4 weeks	Phenolphthalein tablet	①⑤⑥
Qiao and Tao [16]	58 (29/29)	T: 76.67 ±4.67 C: 76.83 ±4.65	Warm acupuncture	ST25, ST36, ST37, CV6; 30 min each time, once a day, total 20 days	Blank	⑦⑧
Li et al. [17]	93 (47/46)	T: 69.82 ±8.42 C: 70.73 ±8.88	Electroacupuncture	ST25; dilatational wave, 2/15 Hz; 30 min each time, once a day, total 4 weeks	Polyethylene glycol	①②③④⑤⑥
Liu et al. [18]	96 (48/48)	T: 65.28 ±7.32 C: 64.58 ±7.33	Electroacupuncture + Phenolphthalein tablet	ST36, ST25; continuous wave, 1.5 Hz; 20 min each time, once a day, total 7 days	Phenolphthalein tablet	①②③④⑤
Jiang et al. [19]	66 (33/33)	T: 64.82 ±12.23 C: 64.41 ±11.89	Warm acupuncture	ST25, ST36, ST37, CV6; 30 min each time, once a day, total 20 days	Blank	⑦⑧
Li et al. [20]	60 (30/30)	T: 76.77 ±11.50 C: 75.77 ±11.30	Press-needle	CV12, CV10, CV6, CV4, BL20, BL21, BL25, BL27; 4 h each time, twice a week, total 30 days	Mosapride citrate tablet	①②③④⑤⑥⑧
Li et al. [21]	78 (39/39)	T: 63.90 ±7.34 C: 63.74 ±9.24	Manual acupuncture	ST25, CV4, ST37, GV29, GV24, GV21, GV19, next to GV20 1.5 cun bilateral; 30 min each time, 3 times a week, total 4 weeks	Sham acupuncture	⑧⑨
Ling et al. [22]	80 (40/40)	T: 60.24 ±6.55 C: 62.14 ±7.58	Press-needle	CV12, ST25, BL21, BL25, ST37; 6–8 h each time, once every other day, total 7 days	Blank	①

C, control group; T, treatment group; ①, response rate; ②, rectal resting pressure; ③, anal resting pressure; ④, anal maximum contraction pressure; ⑤, rectal sensation threshold; ⑥, rectal maximum tolerable volume; ⑦, the Constipation Scoring System score; ⑧, the Patient Assessment of Constipation Quality of Life score; ⑨, adverse events.

performed a blinding method on subjects and operators, and described a blinding method on outcome assessments [21]. All the studies with case dropouts described detailed reasons. Only one study was conducted according to protocols, so the selective reporting was judged as “low risk” [21].

3.5. Intervention effects

3.5.1. Response rate

Five studies with 389 participants reported response rate [15,17,18,20,22]. Two studies of acupuncture vs. medicine showed a significant difference (RR=1.29, 95 %CI= 1.09 to 1.54, $P = 0.003$), with low heterogeneity ($I^2=0\%$, $P = 0.76$) [17,20]. One study of acupuncture vs. blank showed a significant difference (RR=1.35, 95 %CI= 1.04 to 1.74, $P = 0.02$) [22]. Two studies of acupuncture + medicine vs. medicine showed no significant difference (RR=1.14, 95 %CI= 0.95 to 1.36, $P = 0.16$), with moderate heterogeneity ($I^2=51\%$, $P = 0.16$) [15,18]. For details, please refer to Fig. 3.

3.5.2. Anal rectum dynamics

Three studies measured rectal resting pressure [17,18,20]. Two studies of acupuncture vs. medicine showed no significant difference (MD=2.91 mmHg, 95 %CI= -0.34 to 6.17, $P = 0.08$), with low heterogeneity ($I^2=0\%$, $P = 0.96$) [17,20]. One study of acupuncture + medicine vs. medicine showed a significant difference (MD=2.03 mmHg, 95 %CI= 0.38 to 3.68, $P = 0.02$) [18]. For details, please refer to Fig. 4.

Three studies measured anal resting pressure [17,18,20]. Two studies of acupuncture vs. medicine showed no significant difference (MD=-1.59 mmHg, 95 %CI= -5.09 to 1.90, $P = 0.37$), with low heterogeneity ($I^2=0\%$, $P = 0.80$) [17,20]. One study of acupuncture +

medicine vs. medicine showed a significant difference (MD=2.49 mmHg, 95 %CI= 0.09 to 4.89, $P = 0.04$) [18]. For details, please refer to Fig. 5.

Three studies measured anal maximum contraction pressure [17,18,20]. Two studies of acupuncture vs. medicine showed no significant difference (MD=-1.37 mmHg, 95 %CI= -9.74 to 7.01, $P = 0.75$), with high heterogeneity ($I^2=61\%$, $P = 0.11$) [17,20]. One study of acupuncture + medicine vs. medicine showed a significant difference (MD=11.90 mmHg, 95 %CI= 2.09 to 21.71, $P = 0.02$) [18]. For details, please refer to Fig. 6.

Four studies measured rectal sensation threshold [15,17,18,20]. Two studies of acupuncture vs. medicine showed no significant difference (MD=-0.43 mL, 95 %CI= -2.54 to 1.68, $P = 0.69$), with low heterogeneity ($I^2=0\%$, $P = 0.72$) [17,20]. Two studies of acupuncture + medicine vs. medicine showed a significant difference (MD=-8.78 mL, 95 %CI= -11.14 to -6.42, $P < 0.00001$), with low heterogeneity ($I^2=0\%$, $P = 0.99$) [15,18]. For details, please refer to Fig. 7.

Three studies measured rectal maximum tolerable volume [15,17,20]. Two studies of acupuncture vs. medicine showed no significant difference (MD=-2.20 mL, 95 %CI= -9.24 to 4.85, $P = 0.54$), with low heterogeneity ($I^2=0\%$, $P = 0.99$) [17,20]. One study of acupuncture + medicine vs. medicine showed a significant difference (MD=-9.95 mL, 95 %CI= -14.47 to -5.43, $P < 0.0001$) [15]. For details, please refer to Fig. 8.

3.5.3. CSS score

Two studies of acupuncture vs. blank reported the CSS score [16,19]. The meta-analysis showed a significant difference between treatment and control groups ($n = 124$; MD=-6.53, 95 %CI= -7.58 to -5.48, $P < 0.00001$), with low heterogeneity ($I^2=0\%$, $P = 0.77$). For details, please refer to Fig. 9.

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
Jiang 2020	+	?	+	?	+	?	+
Li 2017	+	+	+	?	+	?	+
Li 2018	+	?	+	?	+	?	+
Li 2023	+	+	+	+	+	+	+
Ling 2022	+	?	+	?	+	?	+
Liu 2018	+	?	+	?	+	?	+
Luo 2019	+	?	+	?	+	?	+
Qiao 2022	?	?	+	?	+	?	+

Fig. 2. Risk of bias summary.

3.5.4. PAC-QOL score

Four studies reported the PAC-QOL score [16,19–21]. One study of acupuncture vs. medicine showed no significant difference (MD=1.64, 95 %CI=-1.01 to 4.29, $P = 0.22$) [20]. One study of acupuncture vs. sham acupuncture showed a significant difference (MD=-7.85, 95 %CI=-13.12 to -2.58, $P = 0.004$) [21]. Two studies of acupuncture vs. blank showed a significant difference (MD=-22.37, 95 %CI=-24.63 to -20.11, $P < 0.00001$), with low heterogeneity ($I^2=0\%$, $P = 0.37$) [16, 19]. For details, please refer to Fig. 10.

3.5.5. AEs

Only one study described AEs [21]. Six patients (15.4 %) in treatment group experienced mild AEs, such as bleeding, subcutaneous hematoma and sharp pain, while none in control group did. No serious AEs happened in either group.

3.6. Publication bias

Because only eight studies were included, the funnel plot was not conducted to assess publication bias.

4. Discussion

4.1. Summary of main findings

As far as current knowledge goes, this represents the inaugural systematic review and meta-analysis with a specific focus on assessing the effect of acupuncture on PDC. In consideration of the differences in pathogenesis and acupoint selection between PDC and ordinary constipation, as well as ineffectiveness of conventional treatments for PDC, it was of great significance to perform this systematic review. It included eight studies with 591 participants. Response rate is commonly used to categorize the improvement of constipation symptoms, thus it has been designated as the primary outcome indicator. The detail response rate is calculated the percentage of patients without inefficacious in a trial. A significant difference was found in response rate when comparing acupuncture with medicine or blank control. While no significant difference was observed when comparing acupuncture + medicine with medicine. The potential reasons may be related to the limited number of included studies and the limited quality of evidence. The meta-analysis also showed a significant difference in the changes in rectal resting pressure, anal resting pressure, anal maximum contraction pressure, rectal sensation threshold and rectal maximum tolerable volume when acupuncture + medicine was compared with medicine; a significant difference in the changes in the CSS score when acupuncture was compared with blank control; and acupuncture did have a significant effect on the PAC-QOL score when compared with sham or blank control. No significant difference was found in the changes in rectal resting pressure, anal resting pressure, anal maximum contraction pressure, rectal sensation threshold, rectal maximum tolerable volume, and the PAC-QOL score when acupuncture was compared with conventional medicine. These findings should be interpreted with caution due to the limited number, low quality, and high risk of bias of the included studies. Additionally, due to a relative dearth of studies reporting AEs, the safety of acupuncture for PDC remains uncertain.

In terms of acupoint selection, our findings indicated that ST25 (*tianshu*), ST36 (*zusanli*) and ST37 (*shangjuxu*) were the most commonly utilized acupoints. The three acupoints are frequently employed in the regulation of gastrointestinal motility. ST25 (*tianshu*) is located in the abdominal region and can be used to treat proximal conditions due to its proximity to the small intestine. Modern medicine has discovered that both the sympathetic nerve of ST25 (*tianshu*) and the gastrointestinal tract are innervated by the same nerve segment [23]. ST36 (*zusanli*) has been demonstrated to be a more efficacious acupoint for treating constipation than the tibial nerve [24]. It could be ascribed to the presence of multiple nerves beneath ST36 (*zusanli*), including the perineal nerve, sciatic nerve, and tibial nerve. Stimulation at ST37 (*shangjuxu*) can ameliorate intestinal motility dysfunction, alleviate gastrointestinal symptoms, and partly restore enteric neuron function [25]. The acupoints that regulate intestinal motility are important, but we should also pay attention to the acupoints for treating neurodegenerative disorders. Li et al. [21] selected some head acupoints of the Governor Vessel, such as GV29 (*yintang*), GV24 (*shenting*), GV21 (*qianding*) and GV19 (*houling*), and then found acupuncture could effectively relieve constipations and improve the quality of life of PD patients, with the treatment effect lasting up to 4 weeks.

4.2. Comparison with other reviews

In recent years, only one meta-analysis has systematically reviewed the effect of acupuncture in treating non-motor symptoms associated with PD [12]. The search strategy failed to assign sufficient priority to constipation as a key term, potentially resulting in the omission of relevant studies. Among the four included RCTs about acupuncture for PDC, two utilized Chinese patent medicine as a control intervention, which were excluded from our review. Limited by the insufficient number of included RCTs, the study did not find a significant difference

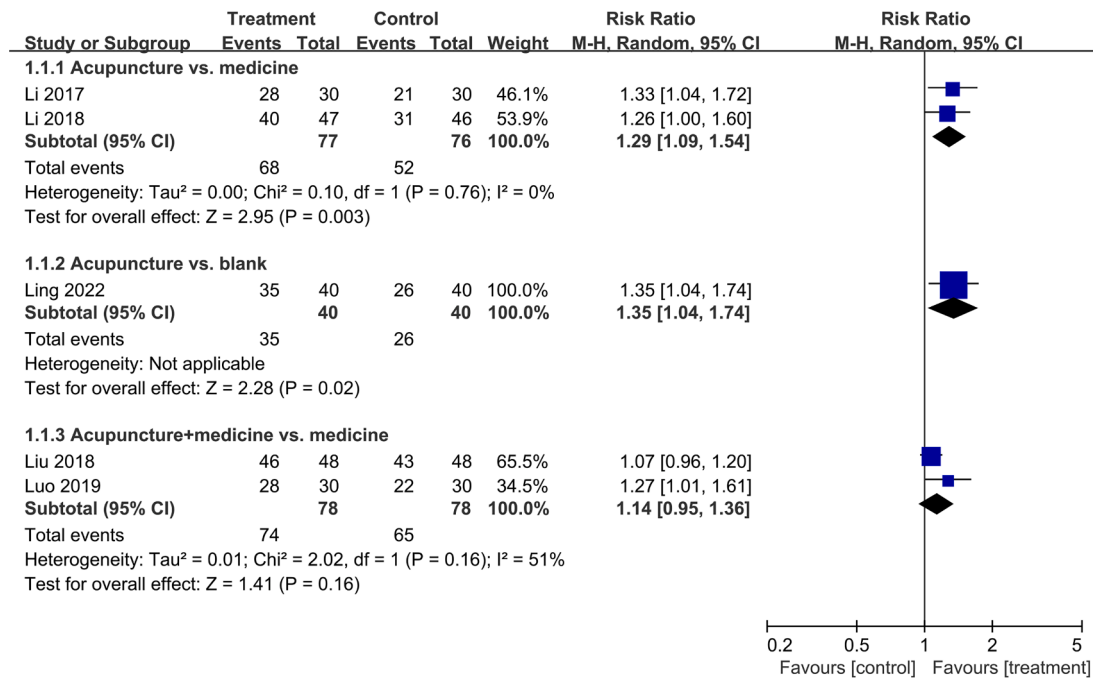


Fig. 3. Forest plot of response rate.

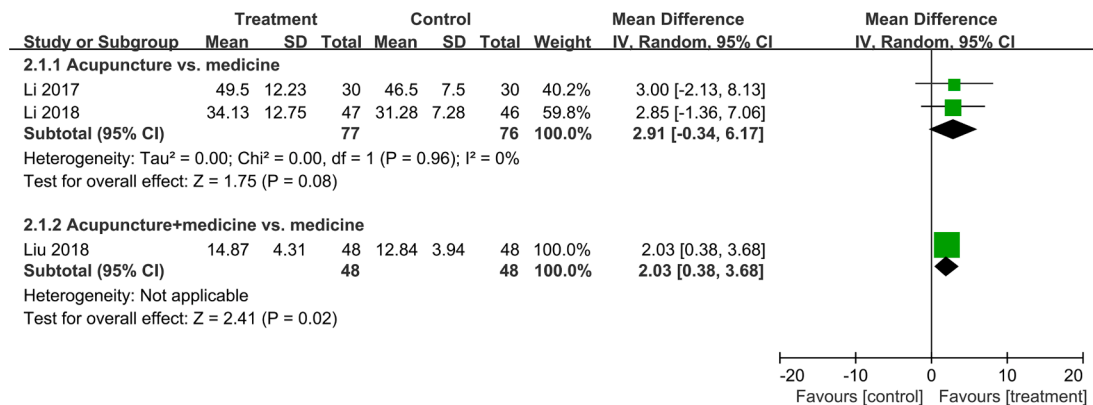


Fig. 4. Forest plot of rectal resting pressure.

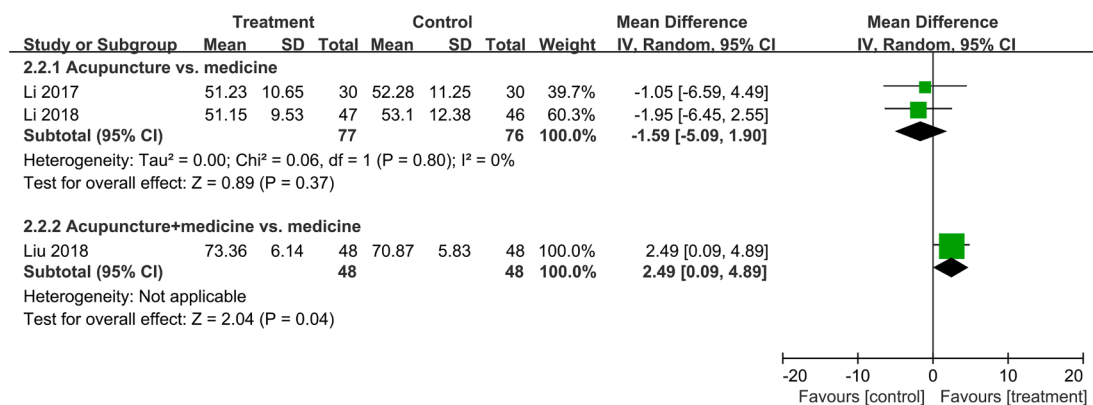


Fig. 5. Forest plot of anal resting pressure.

in terms of PDC, which was not convincing. In our review, the search strategies were meticulously designed, and eight RCTs were included. We performed meta-analysis on various outcomes including response

rate, rectal resting pressure, anal resting pressure, anal maximum contraction pressure, rectal sensation threshold, rectal maximum tolerable volume, CSS score, and PAC-QOL score. Furthermore, we

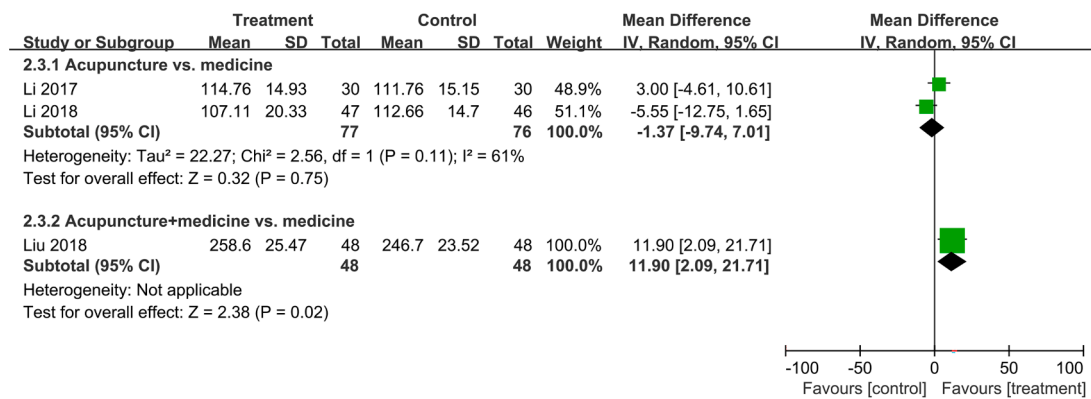


Fig. 6. Forest plot of anal maximum contraction pressure.

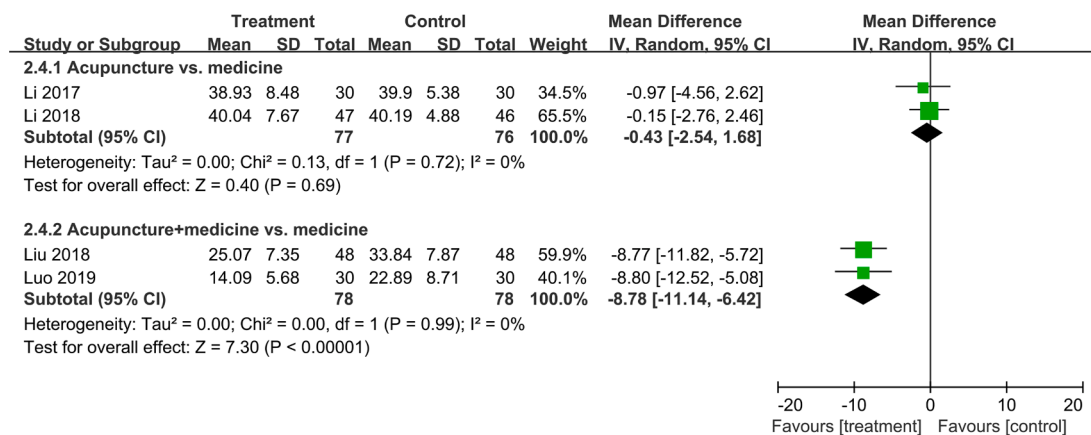


Fig. 7. Forest plot of rectal sensation threshold.

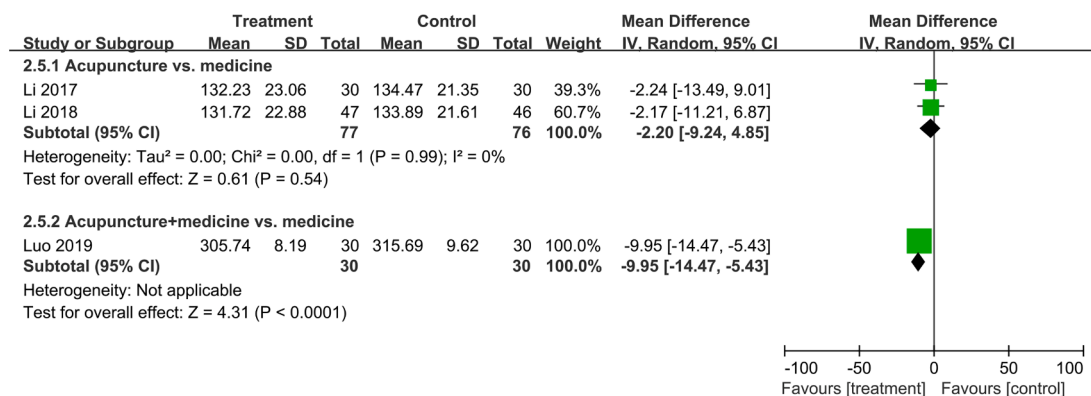


Fig. 8. Forest plot of rectal maximum tolerable volume.

conducted the meta-analysis based on different interventions.

4.3. Mechanisms of acupuncture

The potential mechanism underlying the therapeutic effects of acupuncture on PDC is a topic of great interest. Acupuncture not only effectively relieves constipation, but also exhibits remarkable efficacy in the treatment of brain diseases, rendering it a suitable therapy for PDC. In recent years, the gut-brain axis has emerged as a promising avenue for treating PDC, given its theoretical proposition of a close interplay between the gastrointestinal tract and central nervous system [26]. One study focused on the gut-brain axis in patients with PD using

electroacupuncture [27]. The results showed that scalp-abdominal electroacupuncture could alleviate motor symptoms, improve daily activities, enhance non-motor symptoms such as sleep and fatigue, and alter the composition of gut bacteria at the genus level. Besides, some animal experiments have suggested that the mechanism of acupuncture on PDC may be linked to the promotion of enteric nervous system structural restoration, modulation of Parkin-induced mitochondrial autophagy, and influence on serotonin 4 receptor-mediated cyclic AMP/protein kinase A signaling pathway [28–30].

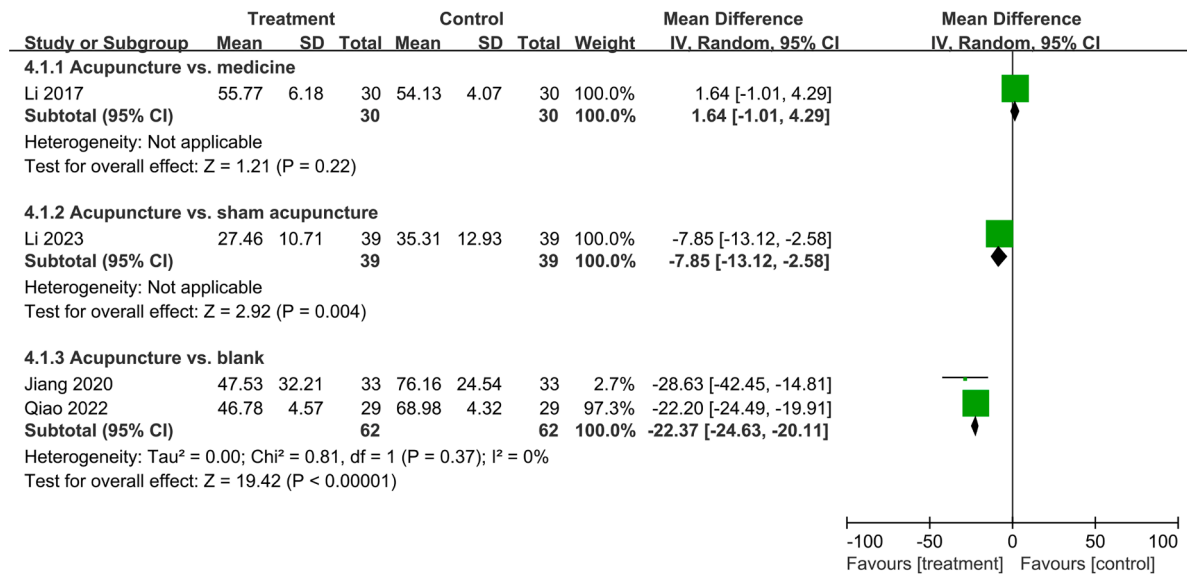


Fig. 9. Forest plot of the CSS score.

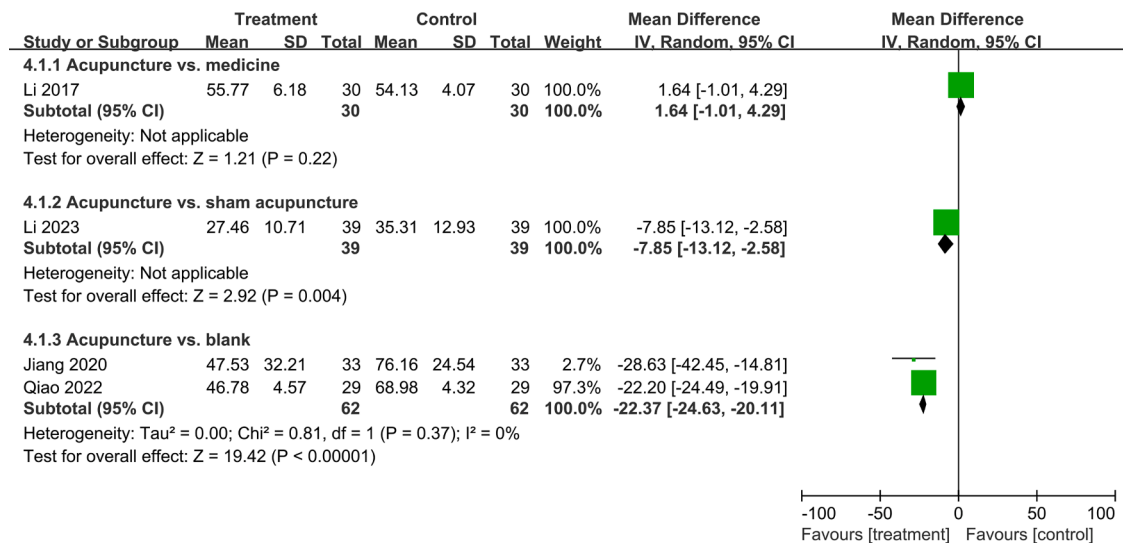


Fig. 10. Forest plot of the PAC-QOL score.

4.4. Strengths and limitations

This study has some strengths. The present systematic review is the first to specifically assess the effect of acupuncture on response rate, rectal resting pressure, anal resting pressure, anal maximum contraction pressure, rectal sensation threshold, rectal maximum tolerable volume, CSS score and PAC-QOL score in PD patients with constipation. The meta-analysis based on different interventions was conducted to fortify the robustness of our conclusions. A standardized protocol was pre-registered in the PROSPERO database, thereby minimizing potential reporting bias and enhancing the credibility of this review. Therefore, our findings may provide reliable evidence to inform clinical treatment decisions and health policies regarding PDC.

However, there are some limitations to this study. Firstly, the scarcity of high-quality RCTs is a critical limitation. Most of the included studies did not emphasize allocation concealment. Only one study was conducted with blinding to prevent performance and detection bias. Most of the included trials were not pre-registered. Secondly, it should be noted that all the studies included in this analysis were conducted

solely within China, which may introduce a potential regional bias. Thirdly, as the number of included RCTs was limited, a funnel plot could not be conducted to assess publication bias. Fourthly, only one study reported AEs related to acupuncture in the treatment of PDC, thus limiting the analysis of safety results.

4.5. Implications

Implications are delineated below. With some limited evidence supporting acupuncture as a potential therapy for PDC and no indication of serious harm, we cautiously recommend it as an adjunctive treatment. To optimize its effects, practitioners may consider using ST25 (*tianshu*), ST36 (*zusanti*) and ST37 (*shangjuxu*). To enhance clinical research quality, future studies should adhere to the Consolidated Standards of Reporting Trials statement and the Standards for Reporting Interventions in Clinical Trials of Acupuncture guideline. Strict enforcement of allocation concealment, blinding, pre-registration and AEs report is imperative. Future research should prioritize the identification of optimal acupuncture parameters, which are critical for advancing the

development of acupuncture. Furthermore, it is necessary to conduct high-caliber and multicenter RCTs with large sample sizes.

5. Conclusions

Acupuncture might be an effective treatment for PDC. The findings should be interpreted with caution due to the limited quality of evidence. High-caliber and multi-center RCTs with large sample sizes should be carried out.

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Data availability

The data that support the findings of this study are available within the article and its supplementary material.

CRedit authorship contribution statement

Xin-Xin Feng: Methodology, Formal analysis, Investigation, Data curation, Writing – original draft. **Lei Chen:** Methodology, Formal analysis, Investigation, Data curation, Funding acquisition. **Yong-Yi Xu:** Investigation, Data curation, Writing – review & editing. **Kai Zhou:** Resources, Writing – review & editing, Funding acquisition. **Kai-Yu Huang:** Conceptualization, Validation, Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.eujim.2024.102332](https://doi.org/10.1016/j.eujim.2024.102332).

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